

Moments

Definition

Moments can be defined as the arithmetic mean of various powers of deviation taken from the mean of a distribution. These moments are known as central moments.

The first four moments about arithmetic mean or central moments are defined below.

The first four Central moments	Individual series	Discrete series	Continuous series
μ_1	$\frac{\sum(x-\bar{x})}{n}=0$	$\frac{\sum f(x-\bar{x})}{N}=0$	$\frac{\sum f(mid-\bar{x})}{N}=0$
μ_2	$\frac{\sum(x-\bar{x})^2}{n} = \sigma^2$	$\frac{\sum f(x-\bar{x})^2}{N}$	$\frac{\sum f(mid-\bar{x})^2}{N}$
μ_3	$\frac{\sum(x-\bar{x})^3}{n}$	$\frac{\sum f(x-\bar{x})^3}{N}$	$\frac{\sum f(mid-\bar{x})^3}{N}$
μ_4	$\frac{\sum(x-\bar{x})^4}{n}$	$\frac{\sum f(x-\bar{x})^4}{N}$	$\frac{\sum f(mid-\bar{x})^4}{N}$

The first four raw moments are defined below.

The first four Raw Moments	Individual series $d_1 = X - A$ A- any origin	Discrete series $d_2 = \frac{X-A}{C}$ C-common point	Continuous series $d_3 = \frac{mid-A}{C}$ c-class interval width
μ'_1	$\frac{\sum d_1}{n}$	$\frac{\sum fd_2}{N} \times C$	$\frac{\sum fd_3}{N} \times C$
μ'_2	$\frac{\sum d_1^2}{n}$	$\frac{\sum fd_2^2}{N} \times C^2$	$\frac{\sum fd_3^2}{N} \times C^2$
μ'_3	$\frac{\sum d_1^3}{n}$	$\frac{\sum fd_2^3}{N} \times C^3$	$\frac{\sum fd_3^3}{N} \times C^3$
μ'_4	$\frac{\sum d_1^4}{n}$	$\frac{\sum fd_2^4}{N} \times C^4$	$\frac{\sum fd_3^4}{N} \times C^4$

Relation between moments about arithmetic mean and moments about an origin are given below.

$$\mu_1 = \mu'_1 - \mu'_1 = 0$$

$$\mu_2 = \mu'_2 - (\mu'_1)^2$$

$$\mu_3 = \mu'_3 - 3\mu'_1\mu'_2 + 2(\mu'_1)^3$$

$$\mu_4 = \mu'_4 - 4\mu'_3\mu'_1 + 6\mu'_2(\mu'_1)^2 - 3(\mu'_1)^4$$